MILITARY SPECIFICATION

WAVEGUIDE ASSEMBLIES (3.5 to 8.2 GHz)

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments of the Department of Defense.

1. SCOPE

1.1 This specification covers the detail requirements for a series of waveguide assemblies operating over the frequency range of 3.5 to 8.2 GHz.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-7105	-Pipe Threads, Taper, Aeronautical National Form, Symbol Anpt, General Requirements For
MIL-W-87105	-Waveguide Assemblies, General Specification For

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General specification applicability. All requirements of MIL-W-87105 shall apply to this specification with modifications as specified herein. Modifications (options, specific values, exceptions, or additions) to MIL-W-87105 shall be identified by similarity of paragraph headings herein. When the general specification and this document conflict, the requirements of this document shall govern.

FSC 5985

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: ASD/ENESS, Wright-Patterson AFB, OH 45433 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.2 Qualification. This specification requires qualification inspection.

3.3 Parts, materials, and processes

3.3.1.1 Dimensions. All dimensions specified on the figures herein shall apply after plating.

3.3.1.2 Sharp edges. All sharp edges outlined on the figures herein shall be broken 0.015 inch.

3.3.1.3 Surface roughness. Surface roughness on all machined surfaces that are not indicated on the figures herein shall be 125√.

3.4 Design and construction

3.4.1 Functional subassemblies, form factor, and electrical characteristics. Functional subassemblies, form factor, and electrical characteristics for waveguide assemblies shall be in accordance with table I herein.

3.4.1.1 Attachment provisions for flanges. The location of flange holes shall be as indicated on the figures herein.

3.4.2 Interface design requirements. Flexible assemblies shall be plain-holed.

3.5 Performance

3.5.1 Electrical properties. Specific values for the electrical properties of the waveguide assemblies shall be as specified herein.

3.5.1.1 Frequency range. The waveguide assemblies shall operate over the frequency range of 3.5 to 8.2 GHz.

3.5.1.2 VSWR. The VSWR of waveguide assemblies shall not exceed the values specified in table I herein when measured at the frequencies specified in MIL-W-87105.

3.5.1.3 Insertion loss. The insertion loss of waveguide assemblies shall not exceed the values specified in table I herein when measured at the frequencies of 3.5, 4.0, 4.5, 5.0, 5.5, 6.0, 6.5, 7.0, 7.5, 8.0, and 8.2 GHz.

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TABLE I. Design and construction parameters.

WAVEGUIDE ASSEMBLIES	FORM FACTOR	VSWR VALUES	INSERTION LOSS VALUES
DOUBLE RIDGED, RIGID ASSEMBLY	FIGURE 1	1.08:1	0.02 dB/ft
DOUBLE RIDGED, FLEXIBLE ASSEMBLY	FIGURE 2	1.15:1	0.10 dB/ft
DOUBLE RIDGED, E PLANE ELBOW - 90°	FIGURE 3	1.08:1	0.05 dB
DOUBLE RIDGED, E PLANE ELBOW -45° (1.766 inches)	FIGURE 4	1.08:1	0.05 dB
DOUBLE RIDGED, E PLANE ELBOW -45° (2.50 inches)	FIGURE 5	1.08:1	0.05 dB
DOUBLE RIDGED, H PLANE ELBOW - 90°	FIGURE 6	1.08:1	0.05 dB
DOUBLE RIDGED, H PLANE ELBOW - 45°	FIGURE 7	1.08:1	0.05 dB
DOUBLE RIDGED, STRAIGHT, 90° TWIST	FIGURE 8	1.10:1	0.06 dB
ADAPTER-WAVEGUIDE, PRESSURE WINDOW	FIGURE 9	1.08:1	0.05 dB
ADAPTER-WAVEGUIDE, PRESSURE WINDOW, PLAIN HOLE	FIGURE 10	1.08:1	0.05 dB
ADAPTER-WAVEGUIDE, PRESSURIZED, PRESSURE WINDOW, PLAIN HOLE	FIGURE 11	1.08:1	0.05 dB
ADAPTER-WAVEGUIDE, PRESSURIZED, PLAIN HOLE	FIGURE 12	1.08:1	0.05 dB
FLANGE-WAVEGUIDE, PRESSURIZED, TAPPED HOLES	FIGURE 13	---	---
FLANGE-WAVEGUIDE, PRESSURIZED, PLAIN HOLE	FIGURE 14	---	---
FLANGE-WAVEGUIDE, TAPPED HOLES	FIGURE 15	---	---
FLANGE-WAVEGUIDE, PLAIN HOLE	FIGURE 16	---	---
GASKET-WAVEGUIDE	FIGURE 17	---	---

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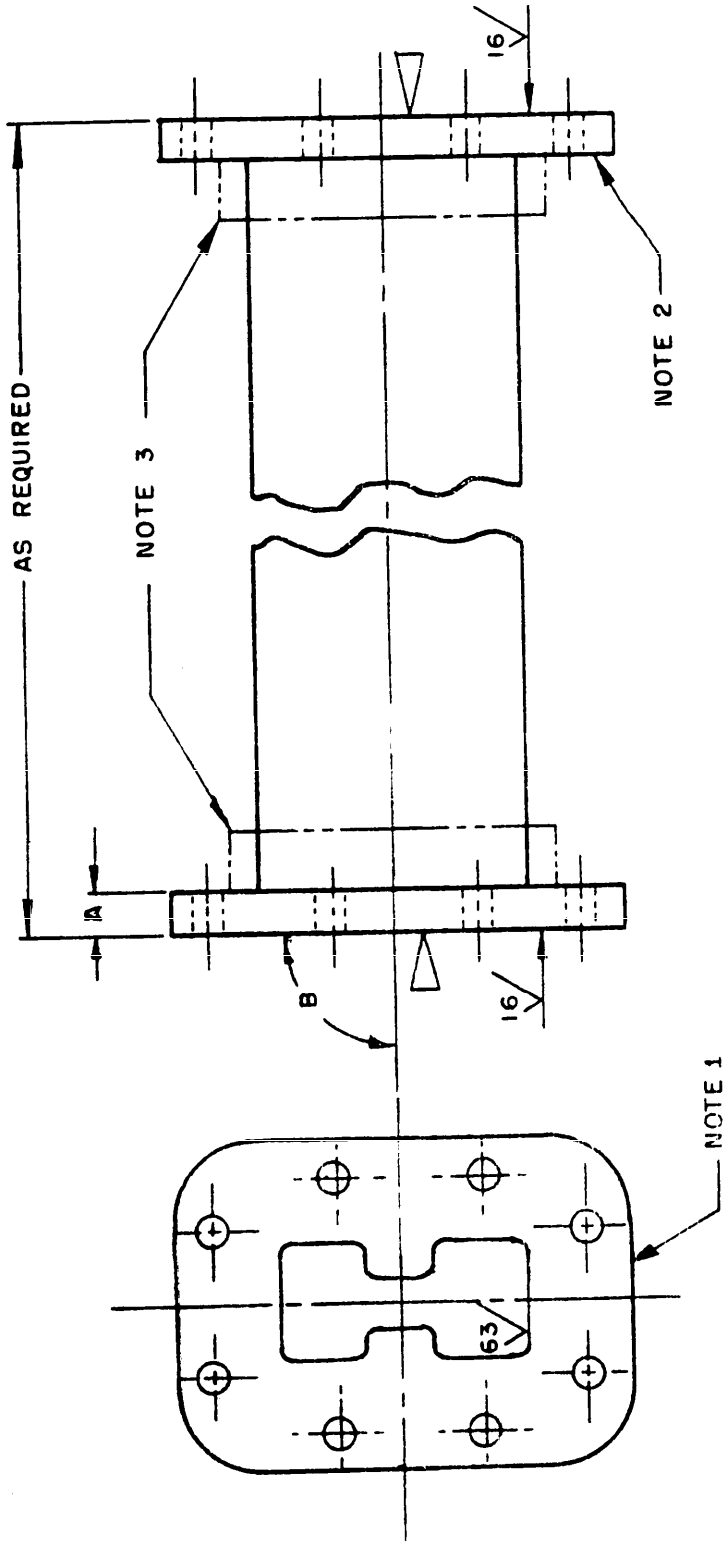


FIGURE 1 (Sheet 1 of 2). Waveguide - double ridged, rigid assembly.

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CODE LETTER	INCHES		MM	
	DIMENSION	TOLERANCE	DIMENSION	TOLERANCE
A	0.250	+0.005 ±0.015	6.35	+0.13 −0.38
B	90°	±15′	—	—

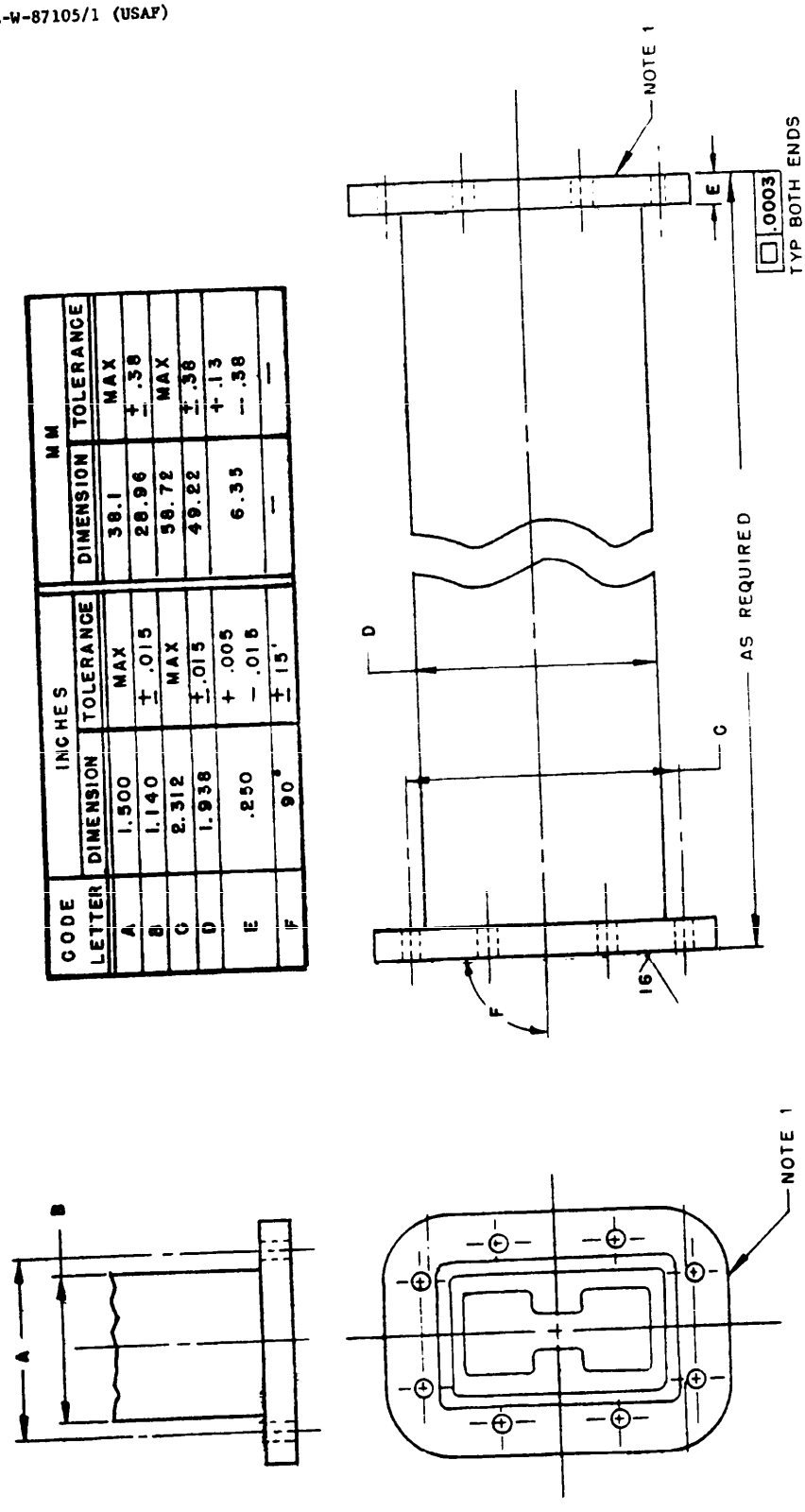
NOTES:

1. Flange face dimensions shall be as specified on figure 14.
2. Flange face dimensions shall be as specified on figure 15.
3. Maximum allowable dimensions for flange buildup shall be as specified on figure 14 or 15.

FIGURE 1 (Sheet 2 of 2). Waveguide - double ridged, rigid assembly.

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CODE LETTER	INCHES		MM	
	DIMENSION	TOLERANCE	DIMENSION	TOLERANCE
A	1.500	MAX	38.1	MAX
B	1.140	± .015	28.96	± .38
C	2.312	MAX	58.72	MAX
D	1.938	±.015	49.22	±.38
E	.250	+ .005 − .015	6.35	+ .13 − .38
F	90°	± 15'	—	—

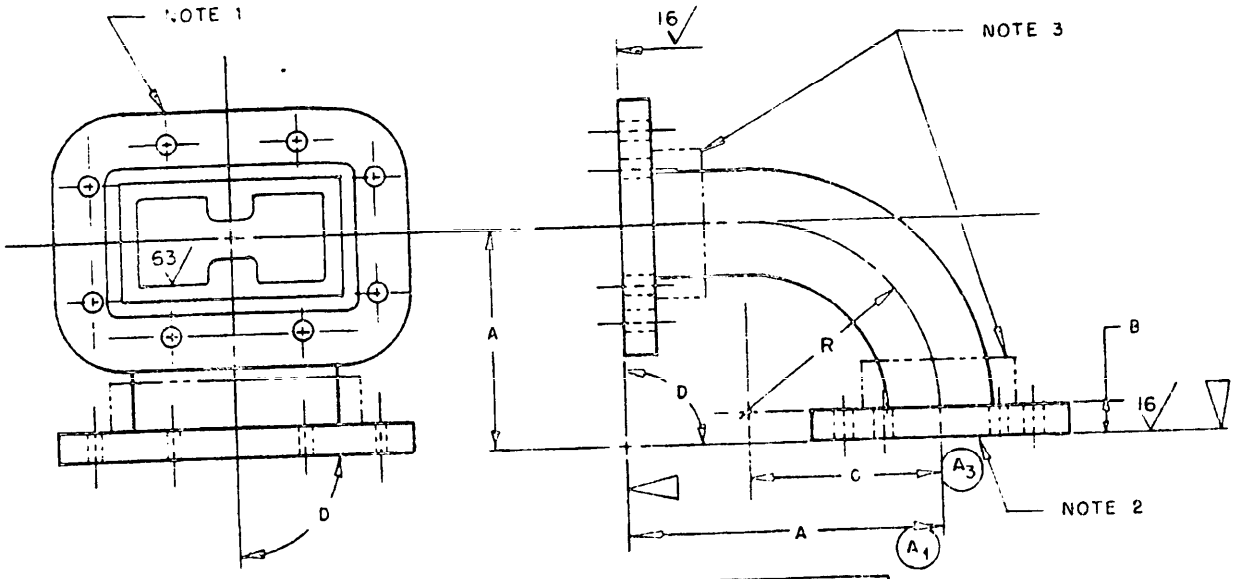


NOTES:

1. Flange face dimensions shall be as specified on figure 14.

FIGURE 2. Waveguide - double ridged, flexible assembly.

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CODE LETTER	INCHES		MM	
	DIMENSION	TOLERANCE	DIMENSION	TOLERANCE
A	1.728	±.020	43.89	±.508
B	.250	+.005 −.015	6.35	+.127 −.381
C	1.478	±.020	37.54	±.508
D	90°	±15'	- -	- -

NOTES:

1. Flange face dimensions shall be as specified on figure 14.
2. Flange face dimensions shall be as specified on figure 15.
3. Maximum allowable dimensions for flange buildup shall be as specified on figure 14 or 15.

FIGURE 3. Waveguide - double ridged, E plane elbow - 90°.

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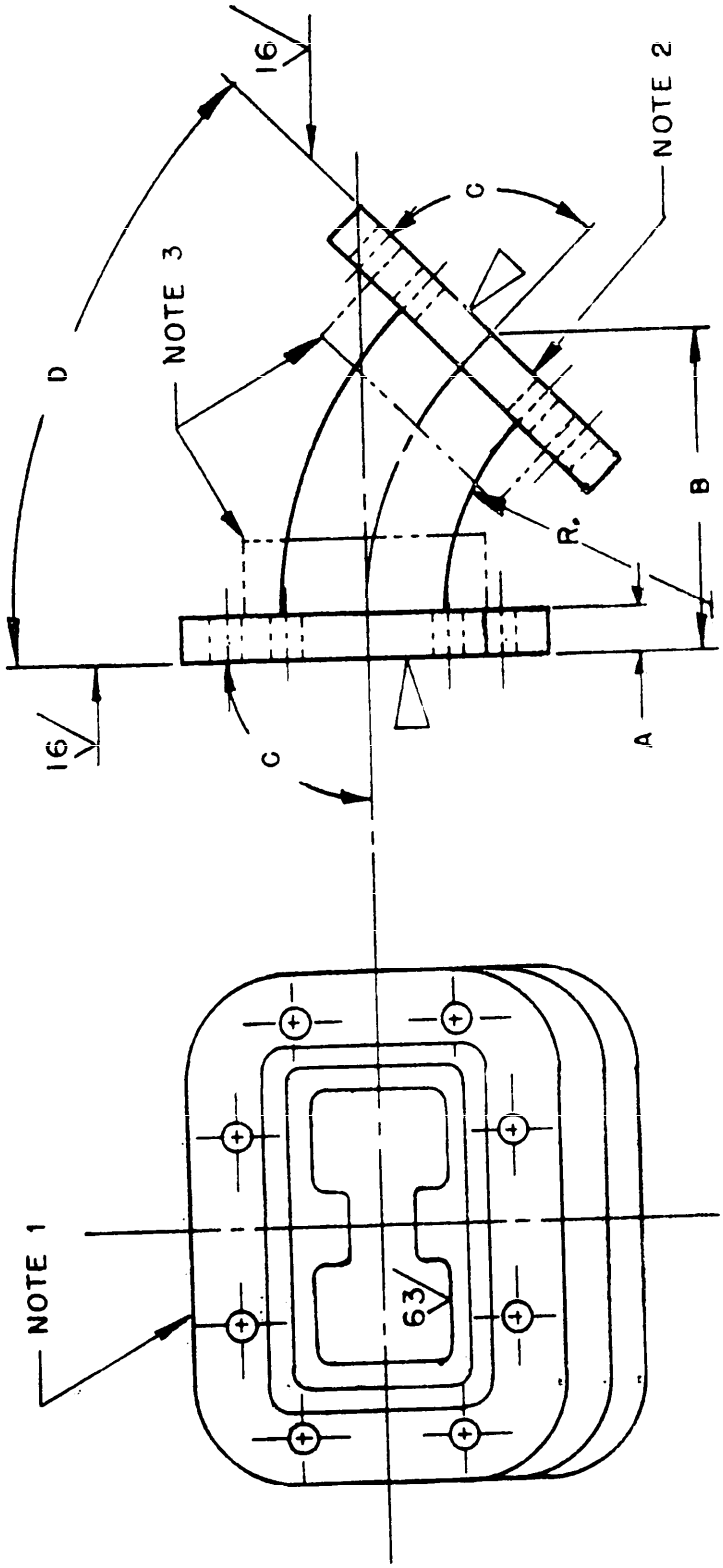


FIGURE 4 (Sheet 1 of 2). Waveguide - double ridged, E plane elbow - 45° (1.766 inches).

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CODE LETTER	INCHES		MM	
	DIMENSION	TOLERANCE	DIMENSION	TOLERANCE
A	.250	+.005 -.015	6.35	+.127 -.381
B	1.776	±.020	44.86	±.508
C	90°	±15'		
D	45°	±30'		

NOTES:

1. Flange face dimensions shall be as specified on figure 14.
2. Flange face dimensions shall be as specified on figure 15.
3. Maximum allowable dimensions for flange buildup shall be as specified on figure 14 or 15.

FIGURE 4 (Sheet 2 of 2). Waveguide - double ridged, E plane elbow - 45° (1.766 inches).

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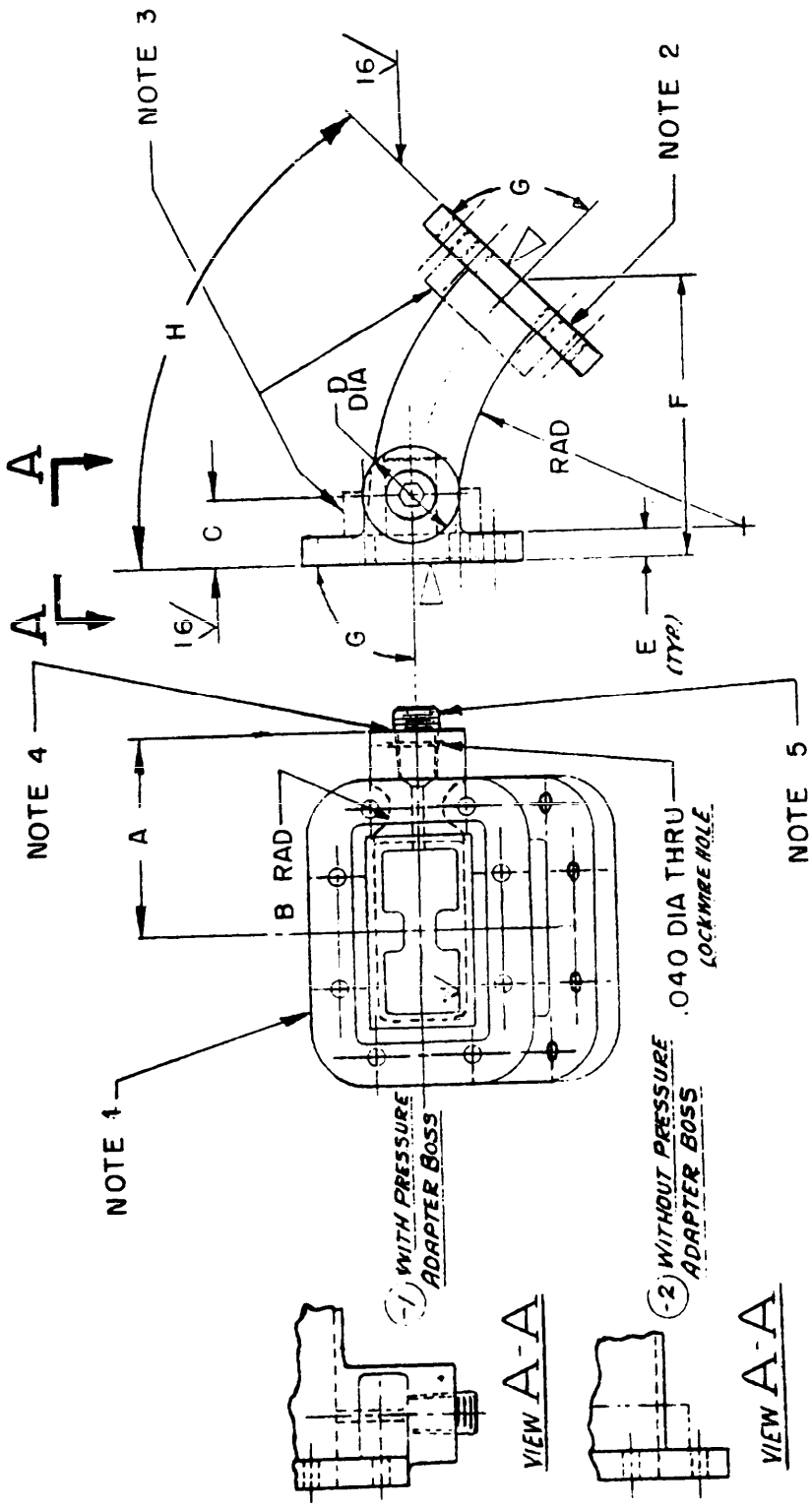


FIGURE 5 (Sheet 1 of 2). Waveguide - double ridged, E plane elbow - 45° (2.50 inches).

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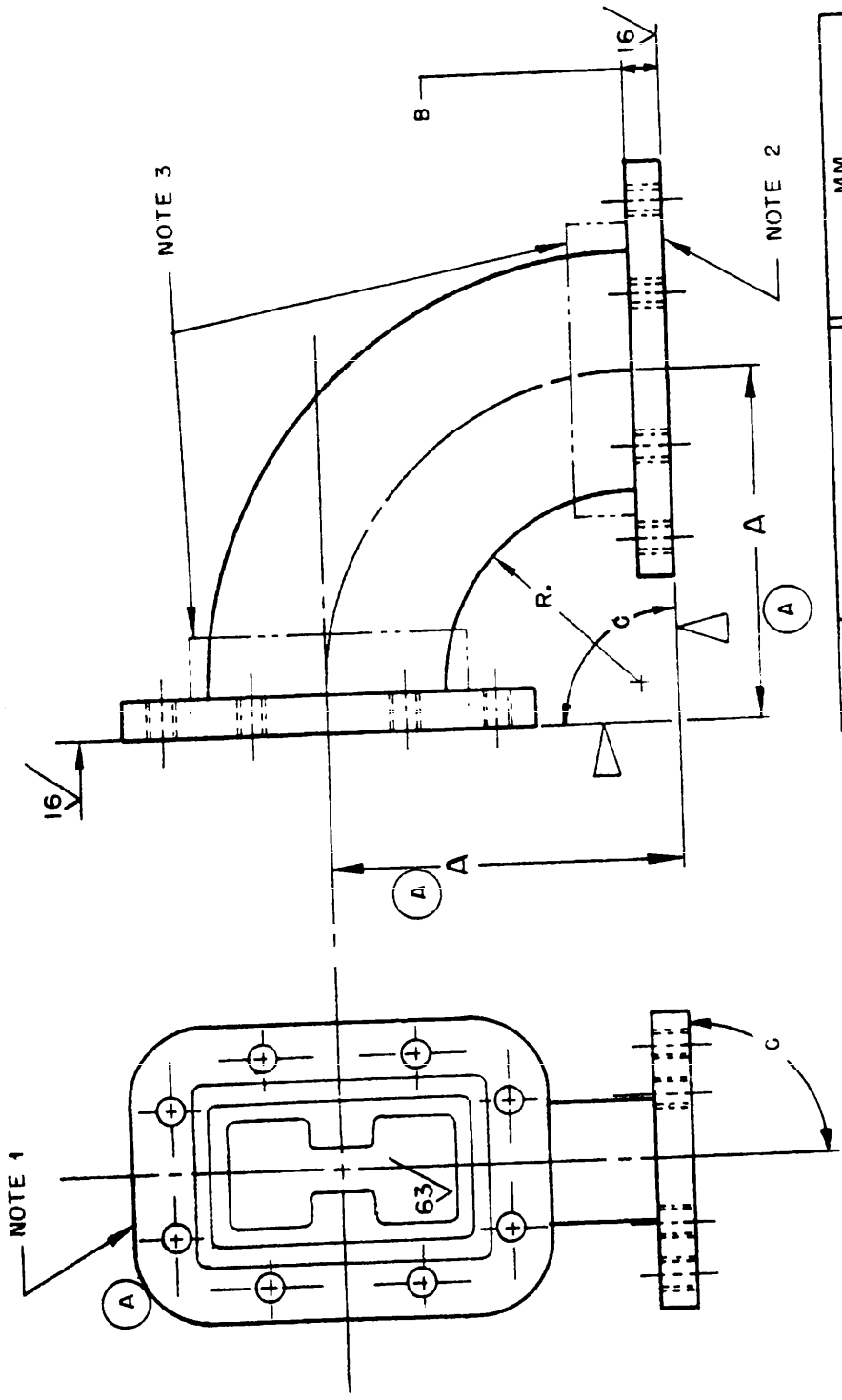
CODE LETTER	INCHES		MM	
	DIMENSION	TOLERANCE	DIMENSION	TOLERANCE
A	1.812	±.0156	46.02	±.396
B	.187	±.0156	4.75	±.396
C	.625	±.010	15.87	±.254
D	.875	±.0156	22.22	±.396
E	.250	+.005 −.015	6.35	+.13 −.38
F	2.500	±.020	63.50	±.51
G	90°	±15′	- - -	- - -
H	45°	±30′	- - -	- - -

NOTES:

1. Flange face dimensions shall be as specified on figure 14.
2. Flange face dimensions shall be as specified on figure 15.
3. Maximum allowable dimensions for flange buildup shall be as specified on figure 14 or 15.
4. DIA. 0.110; +0.004, -0.001 thru tap drill 7/16 deep. Countersink 90° to 13/32 DIA. Thread 1/8 Anpt per MIL-P-7105.
5. Pipe plug - 1/8 socket head.

FIGURE 5 (Sheet 2 of 2). Waveguide - double ridged, E plane elbow - 45° (2.50 inches).

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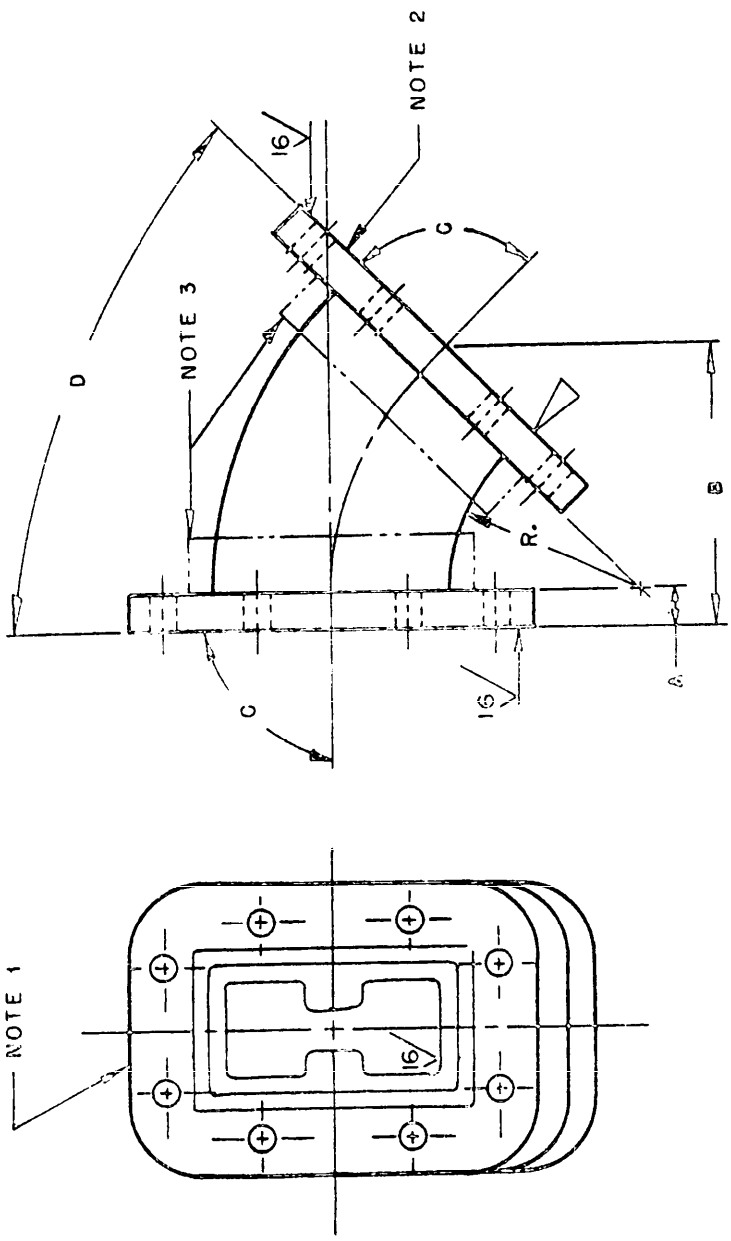
CODE LETTER	INCHES		MM	
	DIMENSION	TOLERANCE	DIMENSION	TOLERANCE
A	2.375	±.020	60.33	±.508
B	.250	+.005 -.015	6.35	+.127 -.381
C	90°	±15'	—	—

NOTES:

1. Flange face dimensions shall be as specified on figure 14.
2. Flange face dimensions shall be as specified on figure 15.
3. Maximum allowable dimensions for flange buildup shall be as specified on figure 14 or 15.

FIGURE 6. Waveguide - double ridged, H plane elbow - 90°.

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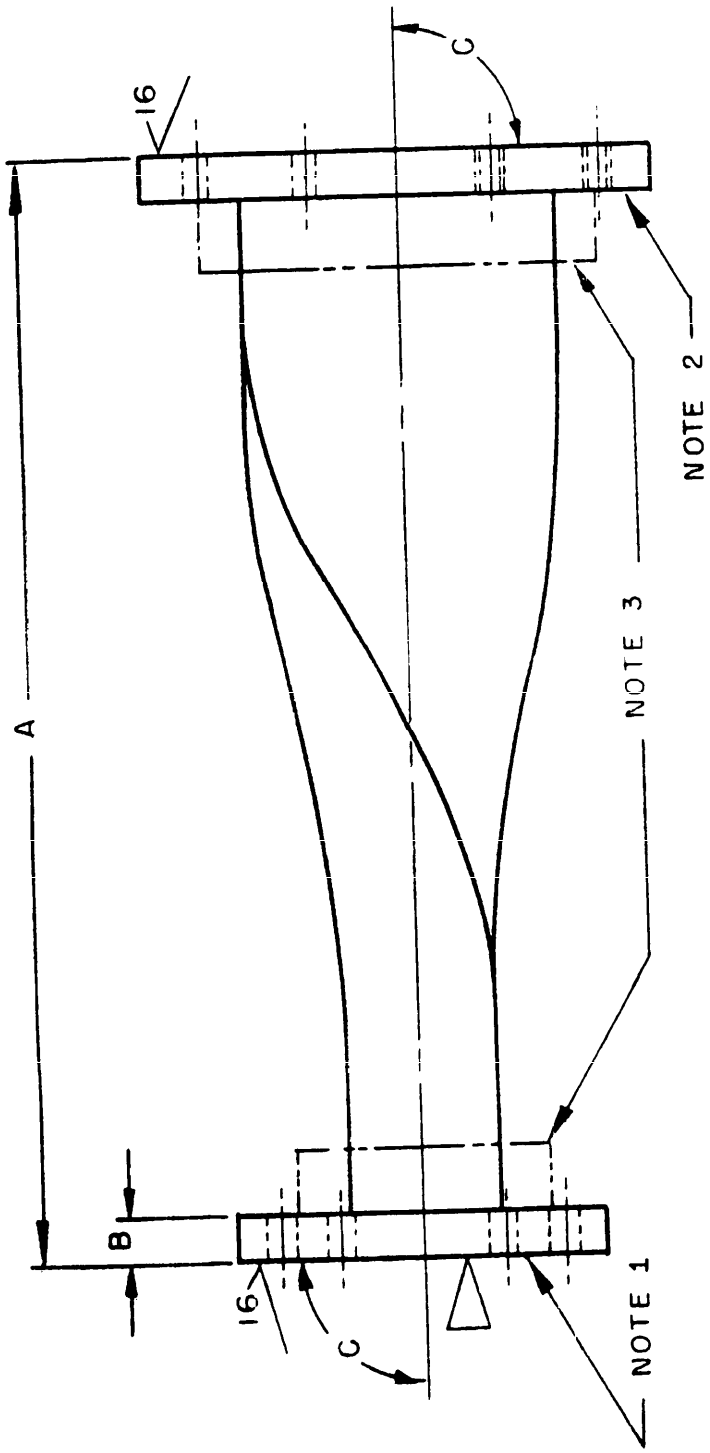
CODE LETTER	INCHES		MM	
	DIMENSION	TOLERANCE	DIMENSION	TOLERANCE
A	.250	+.005 −.015	6.35	+.13 −.39
B	1.938	±.020	49.22	±.51
C	90°	± 15'	—	—
D	45°	± 30'	—	—

NOTES:

1. Flange face dimensions shall be as specified on figure 14.
2. Flange face dimensions shall be as specified on figure 15.
3. Maximum allowable dimensions for flange buildup shall be as specified on figure 14 or 15.

FIGURE 7. Waveguide - double ridged, H plane elbow - 45°.

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CODE LETTER	INCHES		MM	
	DIMENSION	TOLERANCE	DIMENSION	TOLERANCE
A	6.000	±.015	152.4	±.38
B	±.250	+.005 -.015	6.35	+.13 -.38
C	90°	±15	- -	- -

NOTES:

1. Flange face dimensions shall be as specified on figure 14.
2. Flange face dimensions shall be as specified on figure 15.
3. Maximum allowable dimensions for flange buildup shall be as specified on figure 14 or 15.

FIGURE 8. Waveguide - double ridged straight, 90° twist.

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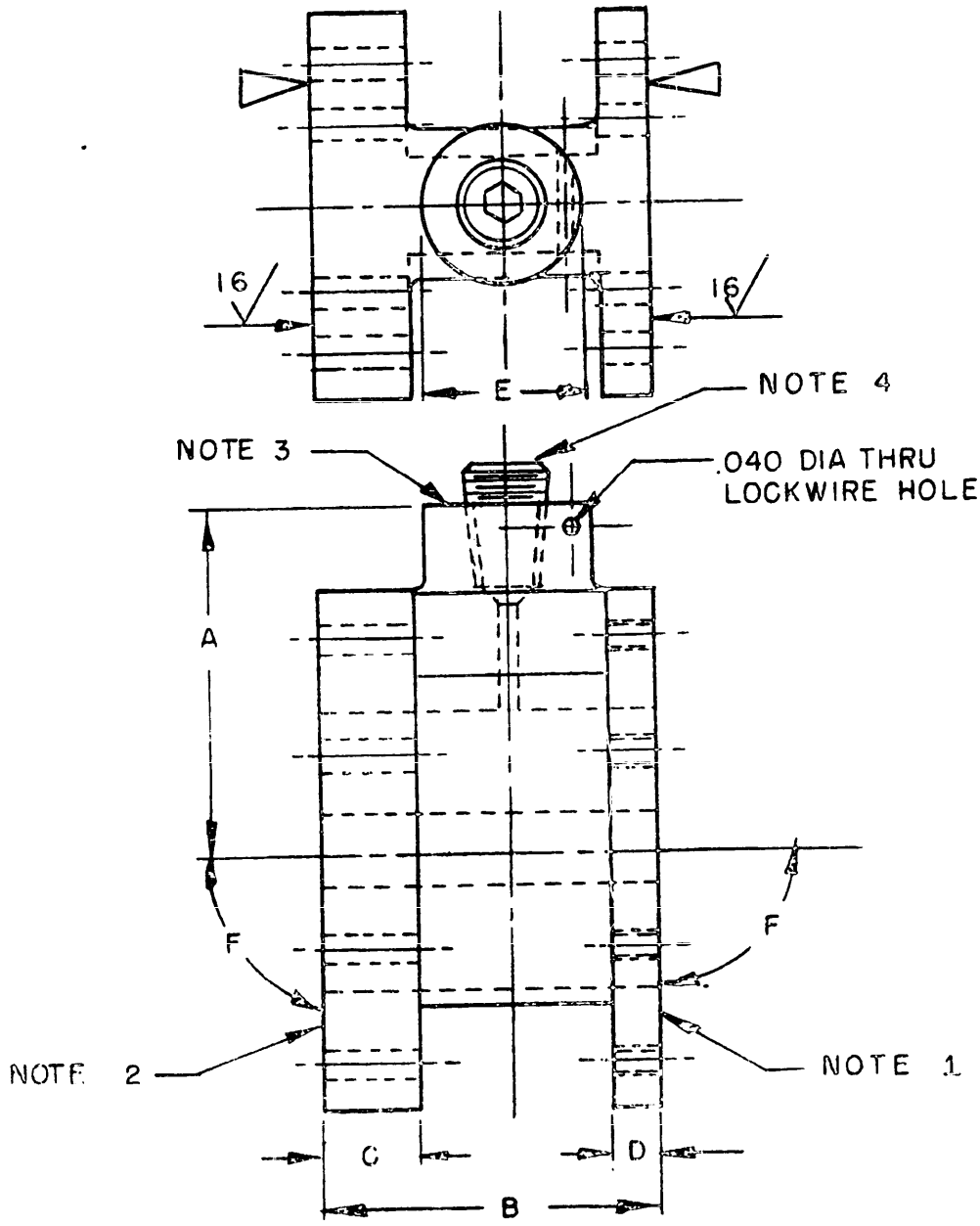


FIGURE 9 (Sheet 1 of 2). Adapter-waveguide, pressure window.

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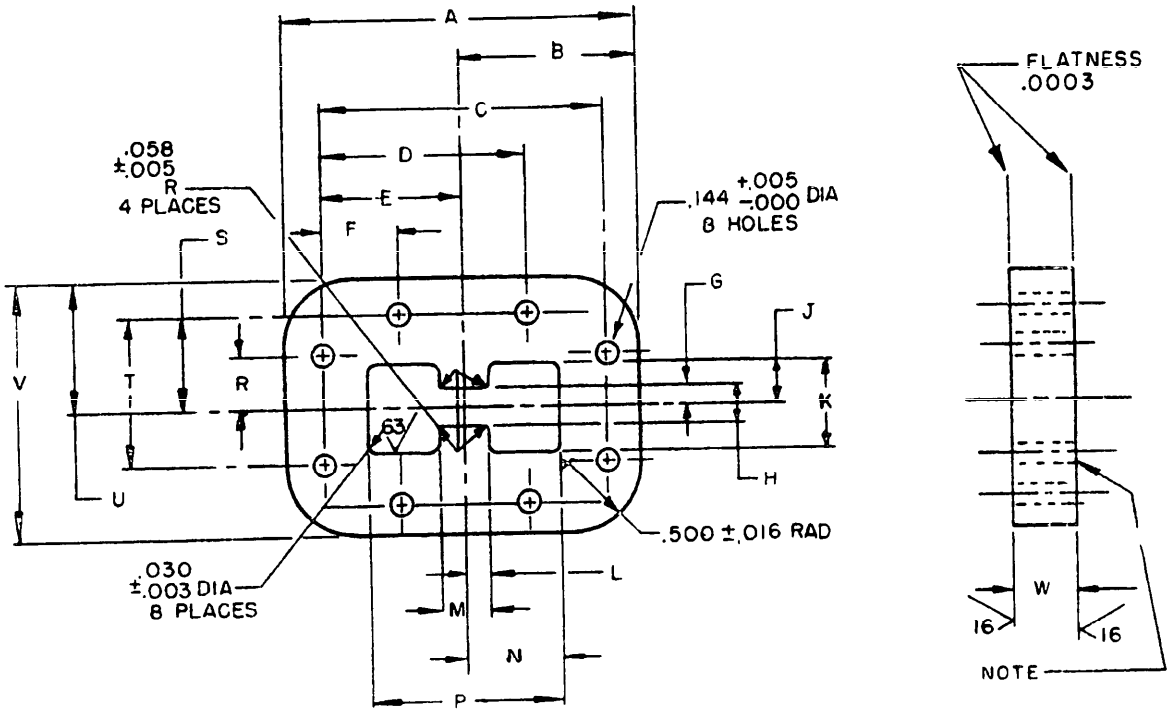
CODE LTR	DIMENSION		TOLERANCE	
	INCHES	MM	INCHES	MM
A	1.812	46.0	MAX	MAX
B	1.750	44.5	+.031 -.000	+0.8 -0.0
C	.500	12.7	+.005 -.015	+.127 -.381
D	.250	6.4	+.005 -.015	+.127 -.381
E	.875	22.2	MAX	MAX
F	.90°	- -	±15'	- -

NOTES:

1. Flange face dimensions shall be as specified on figure 15.
2. Flange face dimensions shall be as specified on figure 16.
3. DIA. 0.110; +0.004, -0.001 thru tap drill 7/16 deep. Countersink 90° to 13/32 DIA. Thread 1/8 Anpt per MIL-P-7105.
4. Pipe plug - 1/8 socket head.
5. Allow clearance for insertion of mounting screws on both sides.

FIGURE 9 (Sheet 2 of 2). Adapter-waveguide, pressure window.

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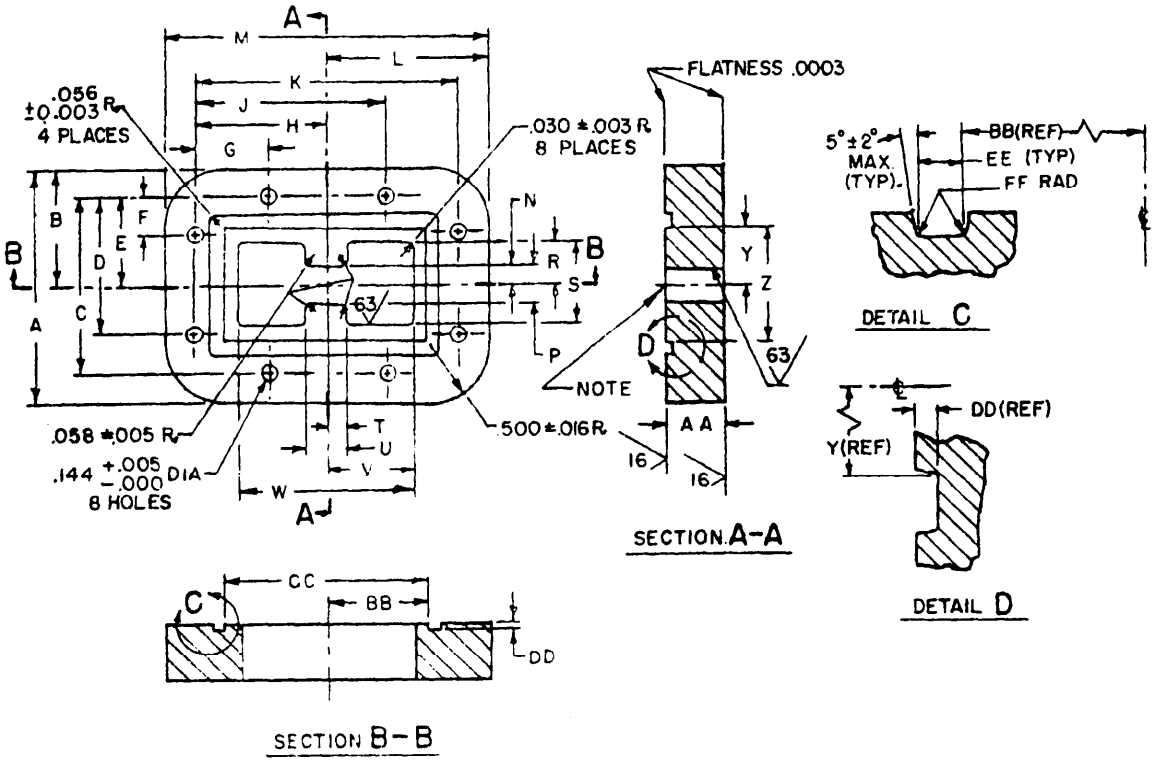


CODE LETTER	INCHES		CODE LETTER	MM	
	DIMENSION	TOLERANCE		DIMENSION	TOLERANCE
A	2.750	±.016	A	69.85	±.41
B	1.375	±.008	B	34.93	±.20
C	2.250	±.002	C	57.15	±.05
D	1.625	±.002	D	41.27	±.05
E	1.125	±.0015	E	28.58	±.04
F	.625	±.002	F	15.87	±.05
G	.146	±.0015	G	3.71	±.04
H	.292	±.003	H	7.42	±.08
J	.344	±.0015	J	8.74	±.04
K	.688	±.003	K	17.47	±.08
L	.185	±.0015	L	4.70	±.04
M	.370	±.003	M	9.40	±.08
N	.740	±.0015	N	18.80	±.04
P	1.480	±.003	P	37.59	±.08
R	.312	±.002	R	7.92	±.05
S	.750	±.0015	S	19.05	±.04
T	1.500	±.002	T	38.1	±.05
U	1.000	±.008	U	25.4	±.20
V	2.000	±.016	V	50.8	±.41
W	.500	+.005 -.015	W	12.7	+.13 -.38

NOTE: Pressure window shall be capable of withstanding the pressure test from either side.

FIGURE 10. Adapter-waveguide, pressure window, plain hole.

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CODE LETTER	INCHES		MM	
	DIMENSION	TOLERANCE	DIMENSION	TOLERANCE
A	2.000	±.016	50.8	±.41
B	1.000	±.008	25.4	±.20
C	1.500	±.002	38.1	±.05
D	1.188	±.002	30.17	±.05
E	.750	±.0015	19.05	±.04
F	.312	±.002	7.92	±.05
G	.625	±.002	15.87	±.05
H	1.125	±.0015	28.57	±.04
J	1.625	±.002	41.27	±.05
K	2.250	±.002	57.15	±.05
L	1.375	±.008	34.92	±.20
M	2.750	±.016	69.85	±.41
N	.146	±.0015	3.71	±.04
P	.292	±.003	7.42	±.08
R	.344	±.0015	8.74	±.04

CODE LETTER	INCHES		MM	
	DIMENSION	TOLERANCE	DIMENSION	TOLERANCE
S	.688	±.003	17.47	±.08
T	.185	±.0015	4.70	±.04
U	.370	±.003	9.40	±.08
V	.740	±.0015	18.80	±.04
W	1.480	±.003	37.59	±.08
Y	.480	±.0015	12.19	±.04
Z	.960	±.003	24.38	±.08
AA	.500	+.005 -.015	12.7	+.13 -.38
BB	.875	±.0015	22.22	±.04
CC	1.750	±.003	44.45	±.08
DD	.048	±.002	1.22	±.05
EE	.112	±.002	2.84	±.05
FF	.010	max	.254	max

NOTE: Pressure window shall be capable of withstanding the pressure test from either side.

FIGURE 11. Adapter-waveguide, pressurized, pressure window, plain hole.

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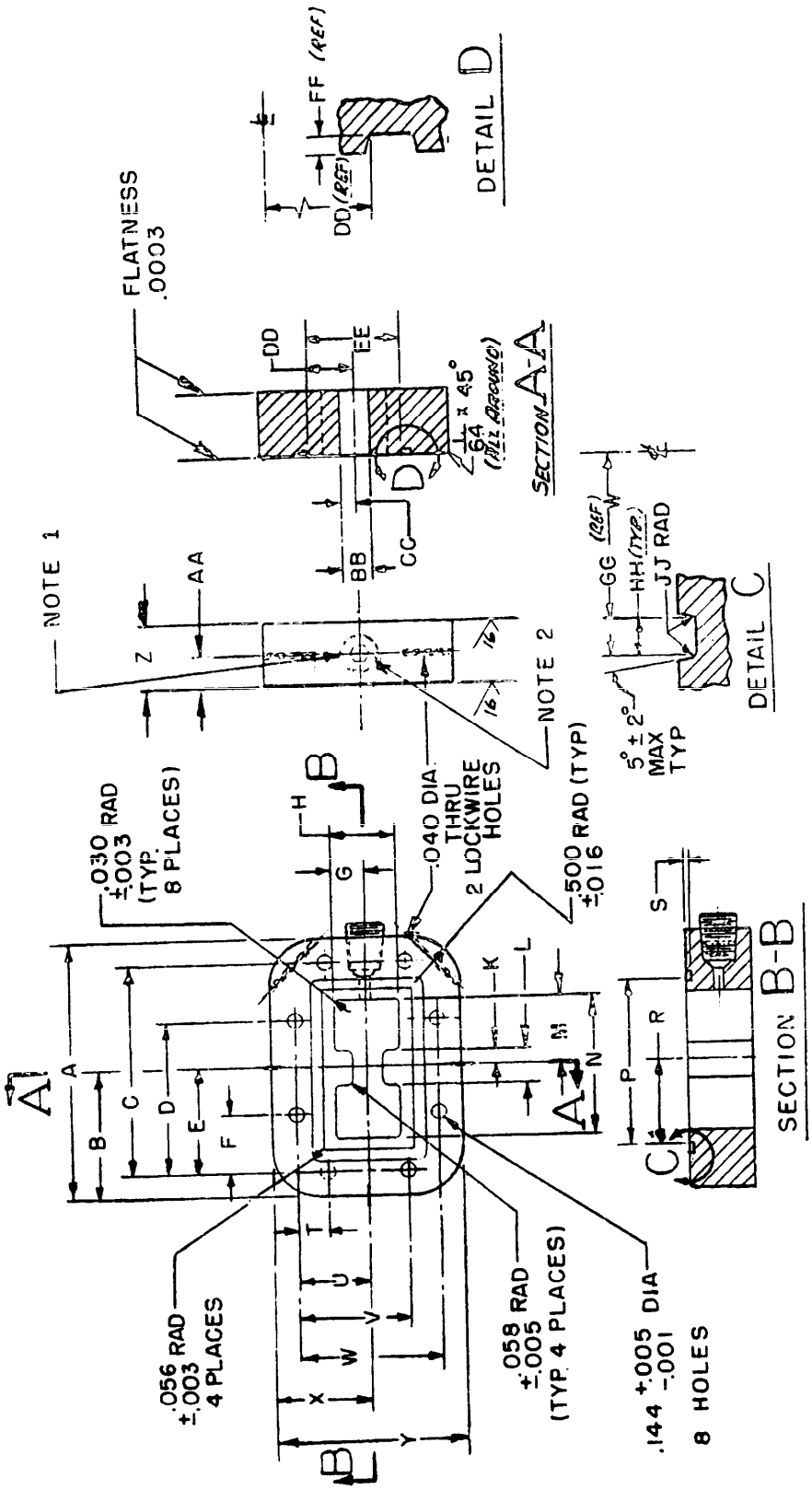


FIGURE 12 (Sheet 1 of 2). Adapter-waveguide, pressurized, plain hole.

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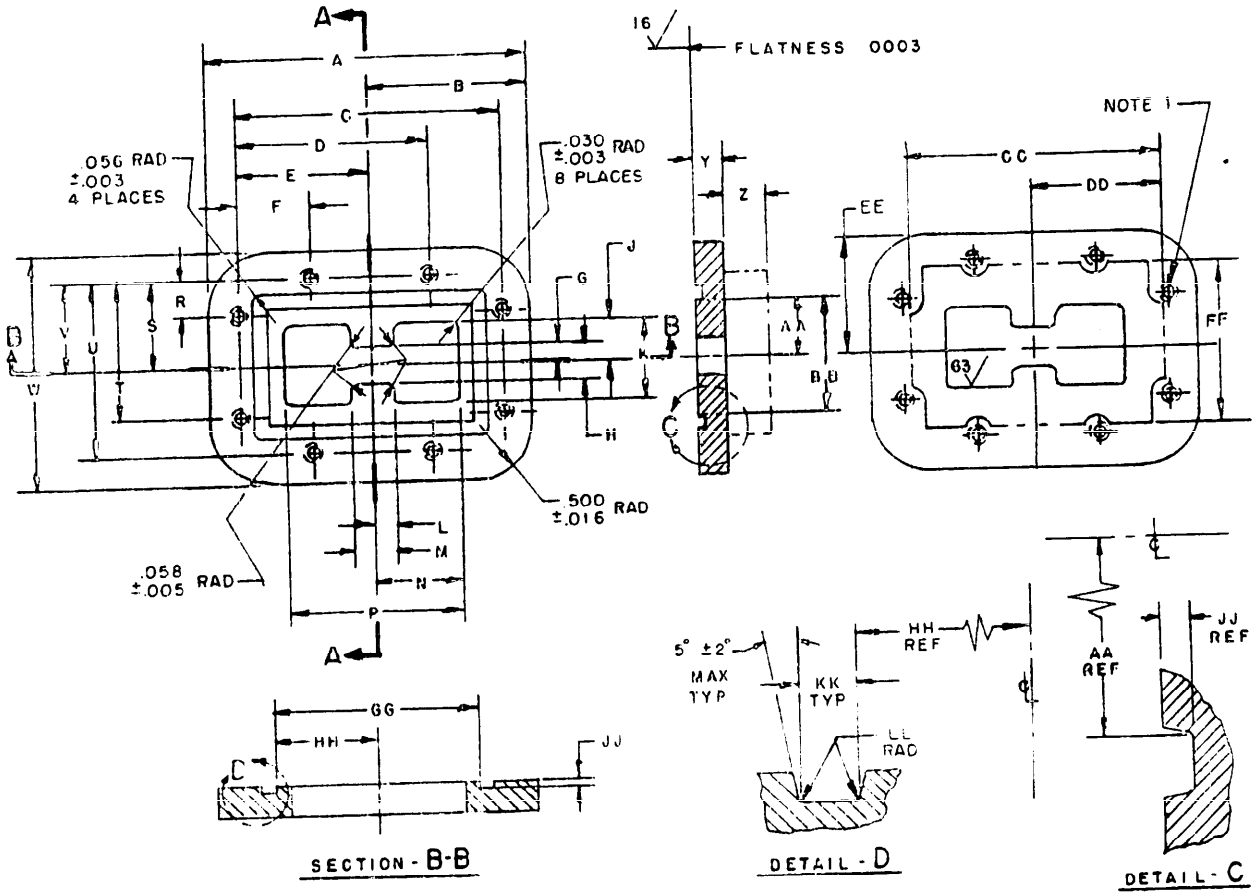
CODE LETTER	INCHES		MM	
	DIMENSION	TOLERANCE	DIMENSION	TOLERANCE
A	2.750	±.016	69.85	±.41
B	1.375	±.008	34.92	±.20
C	2.250	±.002	57.15	±.05
D	1.625	±.002	41.27	±.05
E	1.125	±.0015	28.57	±.04
F	.625	±.002	15.87	±.05
G	.344	±.0015	8.74	±.04
H	.688	±.003	17.47	±.08
K	.185	±.0015	4.70	±.04
L	.370	±.003	9.40	±.08
M	.740	±.0015	18.80	±.04
N	1.480	±.003	37.60	±.08
P	1.750	±.003	44.45	±.08
R	.875	±.002	22.22	±.05
S	.048	±.002	1.22	±.05
T	.312	±.002	7.92	±.05
U	.750	±.0015	19.05	±.04
V	1.188	±.002	30.17	±.05
W	1.500	±.002	38.10	±.05
X	1.000	±.008	25.4	±.20
Y	2.000	±.016	50.8	±.41
Z	.688	±.010	17.47	±.25
AA	.343	±.010	8.71	±.25
BB	.292	±.003	7.42	±.08
CC	.146	±.0015	3.71	±.04
DD	.480	±.002	12.19	±.05
EE	.960	±.003	24.38	±.08
FF	.048	±.002	1.22	±.05
GG	.875	±.002	22.22	±.05
HH	.112	±.002	2.84	±.05
JJ	.010	Max	2.254	Max

NOTES:

1. DIA. 0.110; +0.004, -0.001 thru tap drill 7/16 deep. Countersink 90° to 13/32 DIA. Thread 1/8 Anpt per MIL-P-7105.
2. Pipe plug - 1/8 socket head.

FIGURE 12 (Sheet 2 of 2). Adapter-waveguide, pressurized, plain hole.

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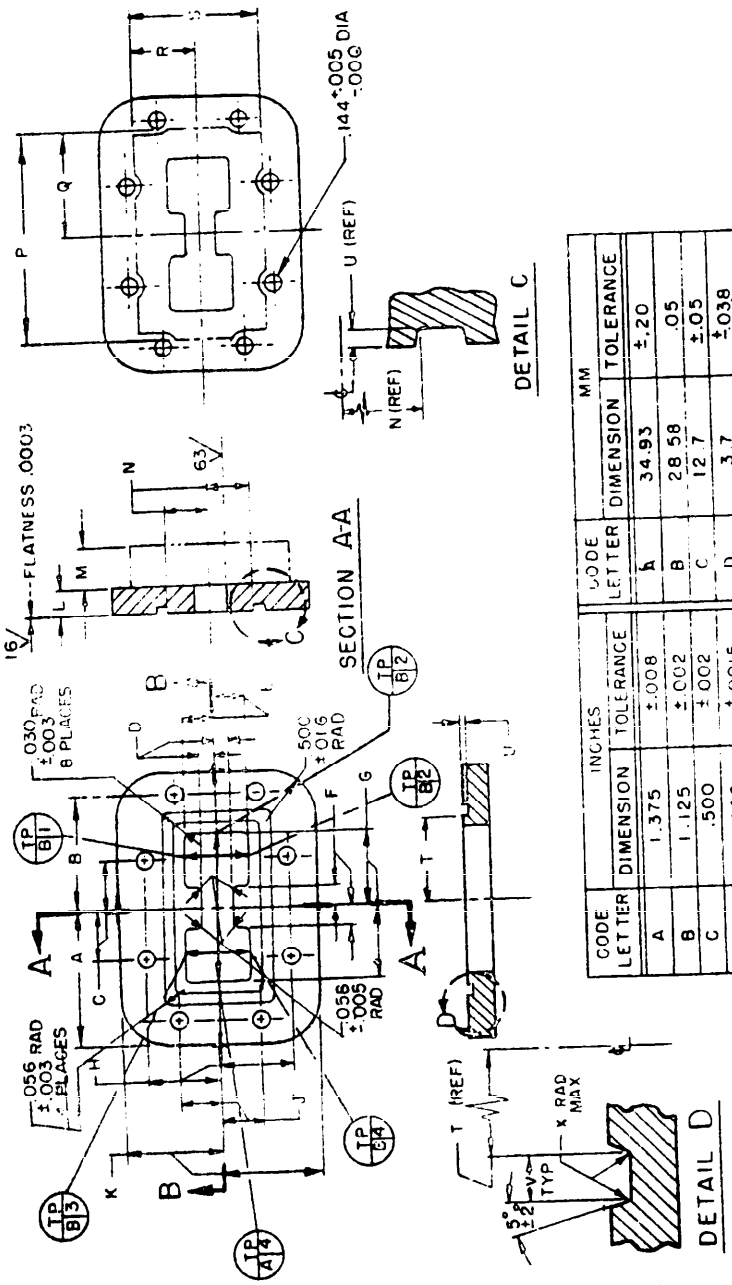
CODE LETTER	INCHES		MM	
A	2.750	± .016	69.85	± .41
B	1.375	± .008	34.92	± .20
C	2.250	± .002	57.15	± .05
D	1.625	± .002	41.27	± .05
E	1.125	± .0015	28.57	± .04
F	.625	± .002	15.87	± .05
G	.146	± .0015	3.71	± .04
H	.292	± .003	7.42	± .08
J	.334	± .0015	8.74	± .04
K	.688	± .003	17.47	± .07
L	.185	± .0015	4.70	± .04
M	.370	± .003	9.40	± .08
N	.740	± .0015	18.80	± .04
P	1.480	± .003	37.59	± .08
R	.312	± .002	7.92	± .05

CODE LETTER	INCHES		MM	
T	1.188	± .002	30.17	± .05
U	1.500	± .002	38.1	± .05
V	1.000	± .008	25.4	± .20
W	2.000	± .016	50.8	± .40
Y	.250	+ .005 − .015	6.35	+ .13 − .38
AA	.480	± .002	12.19	± .05
BB	.960	± .003	24.38	± .08
CC	2.125	MAX	53.97	MAX
DD	1.062	± .010	26.97	± .25
EE	.687	± .010	17.45	± .25
FF	1.375	MAX	34.92	MAX
GG	1.750	± .003	44.45	± .08
HH	.875	± .002	22.22	± .05
JJ	.048	± .002	1.22	± .05
KK	.112	± .002	2.84	± .05
LL	.010	MAX	.254	MAX

NOTE 1: Tap drill thru countersink 110° to 0.138 DIA. (far side). Tap 6- 32NC-2B, 8 holes.

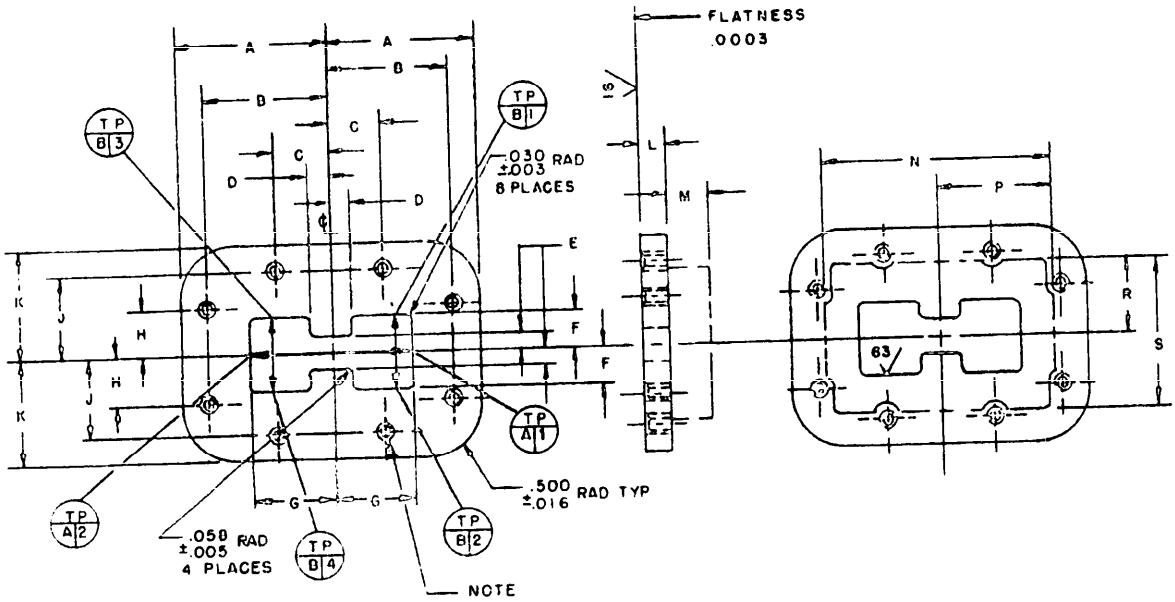
FIGURE 13. Flange-waveguide, pressurized, tapped hole.

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CODE LETTER	INCHES DIMENSION	INCHES TOLERANCE	CODE LETTER	MM DIMENSION	MM TOLERANCE
A	1.375	±.008	A	34.93	±.20
B	1.125	±.002	B	28.58	.05
C	.500	±.002	C	12.7	±.05
D	.146	±.0015	D	3.7	±.038
E	.344		E	8.74	
F	.185	±.0015	F	4.70	±.038
G	.740		G	18.80	
H	.750	±.002	H	19.05	±.05
J	.438	±.002	J	11.13	±.05
K	1.000	±.008	K	25.4	±.20
L	.250	+.005 -.015	L	6.35	+.13 -.38
M	.375	MAX	M	9.52	MAX
N	.480	±.002	M	12.2	±.05
P	2.125	MAX	P	53.97	MAX
Q	1.062	±.010	Q	26.97	±.25
R	.687	±.010	R	17.45	±.25
S	1.375	MAX	S	34.92	MAX
T	.875	±.002	T	22.23	±.05
U	.048	±.002	U	1.22	±.05
V	.112	±.002	V	2.84	±.05
X	.010	MAX	X	.25	MAX

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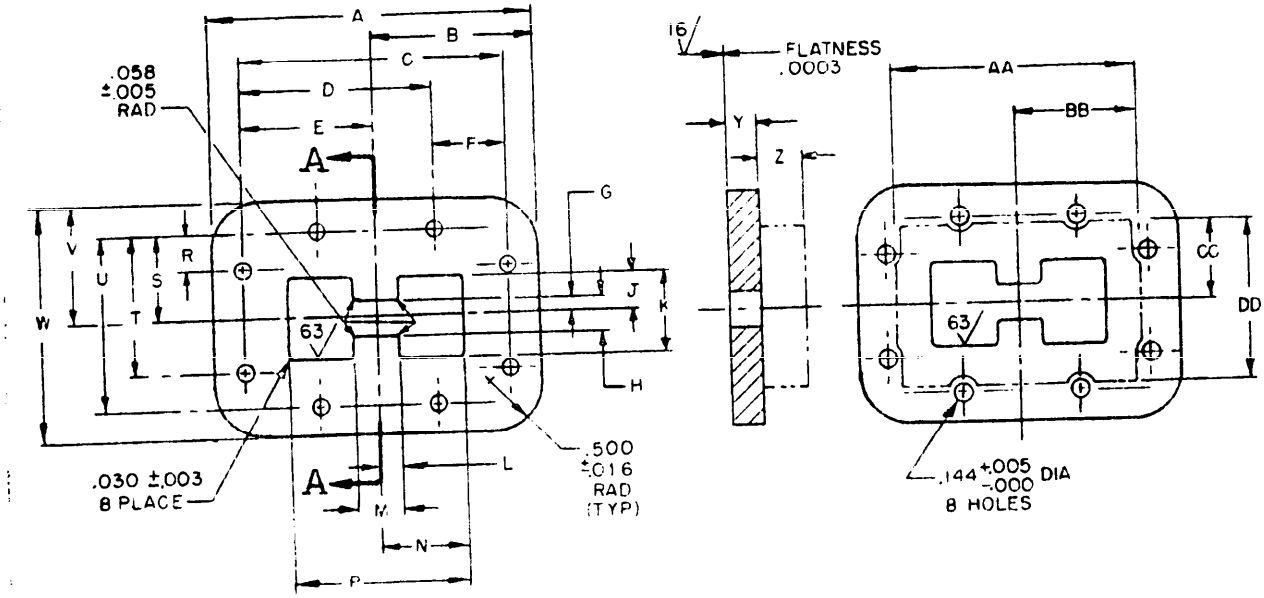


CODE LETTER	INCHES DIMENSION	INCHES TOLERANCE	MM DIMENSION	MM TOLERANCE
A	1.375	± .008	34.92	± .20
B	1.125	± .002	28.75	± .05
C	.500	± .002	12.7	± .05
D	.185	± .0015	4.70	± .04
E	.146	± .0015	3.71	↕
F	.344	± .0015	8.64	
G	.740	± .0015	18.80	± .04
H	.438	± .002	11.13	± .05
J	.750	± .002	19.05	± .05
K	1.000	± .008	25.4	± .20
L	.250	+ .005 — .015	6.35	+ .13 — .38
M	.375	MAX	9.52	MAX
N	2.125	MAX	53.97	MAX
P	1.062	± .010	26.97	± .25
R	.687	± .010	17.45	± .25
S	1.375	MAX	34.92	MAX

NOTE: Tap drill thru countersink 110° to 0.138 DIA. Tap 6- 32UNC-2B, 8 holes.

FIGURE 15. Flange-waveguide, tapped holes.

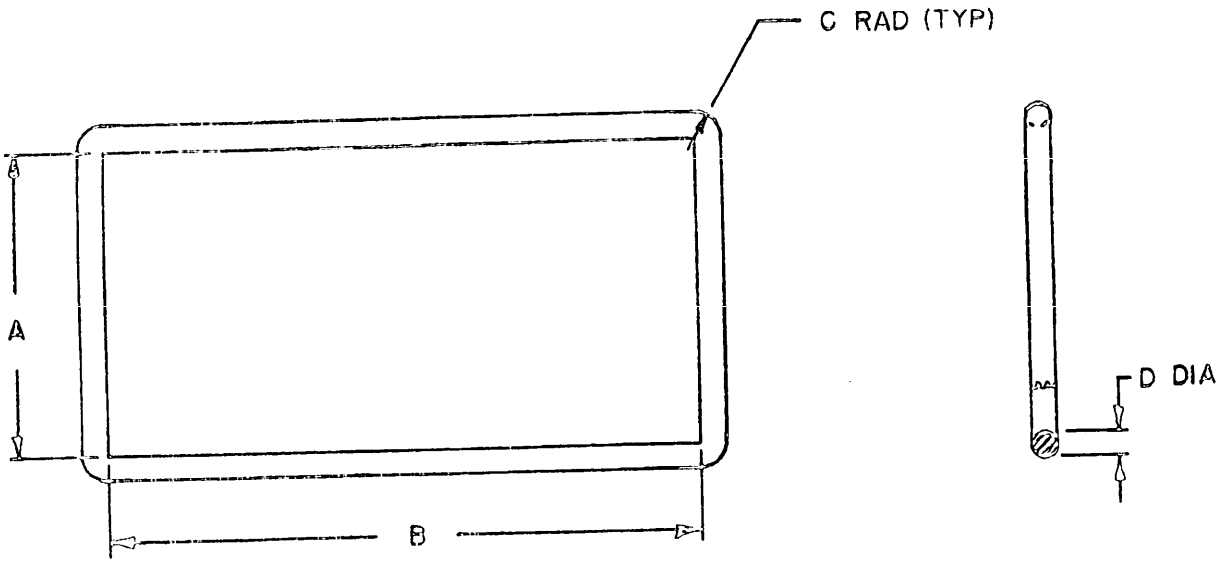
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CODE LETTER	INCHES DIMENSION	INCHES TOLERANCE	MM DIMENSION	MM TOLERANCE
A	.750	±.016	19.05	±.40
B	1.375	±.008	34.93	±.20
C	2.250	±.002	57.15	±.05
D	1.625	±.002	41.27	±.05
E	1.125	±.0015	28.57	±.04
F	.625	±.002	15.87	±.05
G	.146	.0015	3.71	±.04
H	.292	±.003	7.42	±.08
J	.344	±.0015	8.74	±.04
K	.688	±.003	17.47	±.08
L	.185	±.0015	4.70	±.04
M	.370	±.003	9.40	±.08
N	.740	±.0015	18.80	±.04
P	1.480	±.003	37.59	±.08
R	.312	±.002	7.92	±.05
S	.750	±.0015	19.05	±.04
T	1.188	±.002	30.17	±.05
U	1.500	±.002	38.1	±.05
V	1.000	±.008	25.4	±.20
W	2.000	±.016	50.8	±.40
Y	.250	+.005 -.015	6.35	+.13 -.38
Z	.375	MAX	9.52	MAX
AA	2.125	±.002	53.97	±.05
BB	1.062	±.010	26.97	±.25
CC	.687	±.010	17.54	±.25
DD	1.375	MAX	34.92	MAX

FIGURE 16. Flange-waveguide, plain hole.

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CODE LETTER	INCHES		MM	
	DIMENSION	TOLERANCE	DIMENSION	TOLERANCE
A	.992	+.010 -.000	23.42	+.25 -.00
B	1.782	+.010 -.000	45.21	+.25 -.00
C	.078	+.004 -.000	1.98	+.10 -.00
D	.080	±.002	2.03	±.05

FIGURE 17. Gasket-waveguide.

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3.5.1.4 PHC. The waveguide assemblies shall withstand 1000 watts of CW power and 20,000 watts of peak power, at a duty cycle of 5 percent.

3.5.1.5 R-f leakage. The r-f suppression capability of waveguide assemblies shall be as follows:

a. Rigid assemblies: 90 dB when tested as a mated pair

b. Flexible assemblies: 90 dB/ft.

3.5.1.6 Electrical length. The electrical length of waveguide assemblies shall be determined and recorded in accordance with the method specified in MIL-W-87105.

3.5.2 Mechanical requirements. Mechanical values for the waveguide assemblies shall be as specified herein.

3.5.2.1 Interface flatness. Interface flatness, as indicated by on the figures herein, shall be 0.0003.

3.5.2.2 Tensile load stress force

3.5.2.2.1 Rigid assemblies. Rigid assemblies shall withstand a tensile load stress force of 150 pounds.

3.5.2.2.2 Flexible assemblies. Flexible assemblies shall withstand a tensile load stress force of 150 pounds and return to within 0.13 inch of their original configuration.

3.5.2.3 Torque value. Both the rigid assemblies and flexible assemblies shall withstand a torque force of 100 inch-pounds.

3.5.2.4 Minimum bend radius. The minimum bend radius for flexible assemblies shall be 4 inches in the E plane and 6 inches in the H plane.

3.5.2.5 Axial twist. The flexible assemblies shall withstand 200 twists at 30°/ft in both the clockwise and counterclockwise direction, and 10,000 twists at 10°/ft in both the clockwise and counterclockwise direction.

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4. QUALITY ASSURANCE PROVISIONS

4.1 The quality assurance provisions of MIL-W-87105 shall apply to this specification, with additions and exceptions specified herein. When the general specification and this specification conflict, this specification shall govern.

4.2 Quality conformance test program. A waveguide assembly that is procured with lesser frequency coverage than that specified herein shall be subjected to quality conformance testing in the frequency range of 3.5 GHz to the maximum frequency required for that assembly.

4.3 Methods of inspection. The required specific testing values for electrical properties and mechanical requirements shall be as specified herein under 3.5.1 and 3.5.2, respectively.

4.3.1 Insertion loss frequencies. The insertion loss tests shall be conducted at each of the frequencies specified in 3.5.1.3 herein.

4.3.2 Interface flatness test. The interface flatness test shall be conducted in accordance with the requirements of MIL-W-87105.

5. PACKAGING

5.1 Section 5 of MIL-W-87105 shall apply.

6. NOTES

6.1 Section 6 of MIL-W-87105 shall apply.

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